

Common Problems and Solutions

Everyone gets errors in R. Errors are normal.

The important skill is learning how to read the message and try a fix.

Problem 1: File Not Found

What You Might See

```
Error in file(file, "rt") : cannot open the connection
```

or:

```
No such file or directory
```

What It Usually Means

R cannot find your file.

What To Check

Run:

```
getwd()
```

```
[1] "/Users/ylim2/Documents/01_Work/01_MA213/R_onboarding/R_onboarding_repo/R_Onboarding"
```

Then check whether `mydata.csv` is inside that folder.

Also run:

```
file.exists("mydata.csv")
```

```
[1] TRUE
```

If the result is `FALSE`, R cannot find the file.

Possible Fixes

- Move `mydata.csv` into your project folder.
- Check the spelling of the file name.
- Make sure the file ends in `.csv`.
- Make sure you are working in the correct folder.

Common Mistake

`mydata.csv` and `mydata (1).csv` are different file names.

Problem 2: Package Installation Issues

Sometimes your instructor may ask you to install a package.

Example:

```
install.packages("ggplot2")
```

What Can Go Wrong?

- Internet connection problems.
- Typing the package name incorrectly.
- R asks you to restart.
- You do not have permission to install software on the computer.

Possible Fixes

- Check your internet connection.
- Make sure the package name is inside quotation marks.
- Restart RStudio and try again.
- Ask your instructor or campus IT for help if you are on a lab computer.

Tip

You only need to install a package once on your computer.
But you need to load it with `library()` each time you start a new R session.

Problem 3: Typing Mistakes

R is very picky about spelling.

These are all different to R:

```
quiz_score
Quiz_score
quizscore
quiz.score
```

Common Typing Mistakes

Mistake	Example	Fix
Missing quote	<code>read.csv(mydata.csv)</code>	<code>read.csv("mydata.csv")</code>
Misspelled object	<code>quz_score</code>	<code>quiz_score</code>
Missing parenthesis	<code>mean(data\$quiz_score</code>	<code>mean(data\$quiz_score)</code>
Wrong capitalization	<code>Read.csv()</code>	<code>read.csv()</code>

Problem 4: Working Directory Problems

If R is looking in the wrong folder, it may not find your files.

Check the working directory with:

```
getwd()
```

```
[1] "/Users/ylim2/Documents/01_Work/01_MA213/R_onboarding/R_onboarding_repo/R_Onboarding"
```

In RStudio, using an RStudio Project can help keep your working directory organized.

Tip

For each assignment or project, keep your script and data file in the same folder.

Problem 5: Rendering Errors

When rendering a Quarto document, R runs the code from top to bottom.

A document may fail to render if:

- A file is missing.
- A package is not loaded.
- An object is used before it is created.
- There is a typing mistake in a code chunk.

Example

This can fail:

```
mean(data$quiz_score)
```

if `data` was never created earlier in the document.

Fix

Make sure this line appears first:

```
data <- read.csv("mydata.csv")
```

Problem 6: Object Not Found

What You Might See

```
Error: object 'data' not found
```

What It Means

R does not know what `data` is.

Fix

Create the object first:

```
data <- read.csv("mydata.csv")
```

Then run your analysis.

Problem 7: Forgetting Quotation Marks

This is incorrect:

```
read.csv(mydata.csv)
```

This is correct:

```
read.csv("mydata.csv")
```

File names must be in quotation marks.

Troubleshooting Checklist

When something goes wrong, ask:

1. Did I spell everything correctly?
2. Did I use quotation marks around file names?
3. Did I run the earlier lines of code?
4. Is my file in the right folder?
5. Does `file.exists("mydata.csv")` return TRUE?
6. Did I close all parentheses?
7. Did I restart RStudio and try again?

Encouragement

Errors do not mean you are bad at R.

Errors are part of learning. Read the message, try one fix, and run the code again.

Practice: Diagnose the Problem

Practice

For each example, decide what the problem is.

Example 1

```
read.csv(mydata.csv)
```

Question: What is missing?

Example 2

```
mean(data$quiz_score
```

Question: What is missing?

Example 3

```
Data <- read.csv("mydata.csv")  
head(data)
```

Question: Why might this fail?

Answers

1. Example 1 is missing quotation marks around the file name.
2. Example 2 is missing a closing parenthesis.
3. Example 3 may fail because `Data` and `data` are different names in R.

Summary

In this section, you learned how to troubleshoot common problems, including:

- File not found errors.
- Package installation issues.
- Typing mistakes.
- Working directory problems.

- Rendering errors.
- Object not found errors.